

CS-540 Data Literacy & Visualizations

9-1 Submit Project: From Data to Decisions: Exploring and Presenting Meaningful Insights

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Data-Driven Analysis of Public Transit Ridership and Resource Allocation

In today's data-driven world, organizations increasingly rely on data analytics and visualization to guide decision-making processes. The ability to interpret complex datasets and translate them into actionable insights has become a critical competency across industries. Public transportation systems run within highly dynamic environments where decisions must balance operational efficiency, financial sustainability, and community accessibility. Without a strong analytical foundation, these decisions risk being misaligned with actual demand, leading to inefficient resource allocation and reduced service effectiveness.

This project focuses on analyzing public transit ridership and resource allocation across three agencies: King County Metro, Spokane Transit Authority, and the City of Longview. These agencies are three distinct operational scales: large, medium, and small, providing a meaningful basis for comparison. The primary goal of this analysis is to transform raw transit and demographic data into meaningful insights that support both strategic and operational decision-making.

To achieve this, multiple datasets were integrated and analyzed using exploratory data analysis techniques and multi-level data visualization. The findings are presented through a structured data-driven narrative tailored to two distinct stakeholder groups: executive leadership and transit planners or analysts. By aligning insights with stakeholder needs, this project proves how data can be effectively used to support informed decision-making and improve overall system performance.

Data Sources and Relevance

The analysis is built upon two complementary datasets that provide both operational and contextual perspectives on public transit systems. The first dataset includes key transit performance metrics such as annual ridership and operating expenses for each agency (Federal Transit Administration, 2023). These variables are essential for understanding the scale of operations and the financial resources needed to support them.

The second dataset provides population estimates at the county level (U.S. Census Bureau, 2024). Population serves as a critical contextual variable because transit demand is closely tied to the number of people within a service area. By incorporating population data, the analysis moves beyond simple comparisons of total ridership and enables a more meaningful evaluation of service use and efficiency.

The integration of these datasets allows for a more comprehensive understanding of transit performance. Rather than evaluating agencies solely based on raw totals, the analysis considers how effectively each system serves its population. This approach offers deeper insight into the relationship between demand, cost, and operational scale, making the findings more relevant and actionable for decision-makers.

Data Quality Evaluation

Ensuring data quality is a fundamental step in any analytical process, as the reliability of insights depends directly on the accuracy and integrity of the data used. The datasets show strong

accuracy due to standardized government data collection practices (Federal Transit Administration, 2023; U.S. Census Bureau, 2024). In this project, the datasets were evaluated across four key dimensions: accuracy, completeness, consistency, and relevance.

Accuracy is supported by authoritative data sources that follow standardized data collection and reporting practices. This ensures that the values used in the analysis reflect real-world conditions and can be trusted for decision-making purposes. Completeness was confirmed through a detailed inspection of the datasets, which revealed no missing values in the selected variables. This is particularly important because missing data can introduce bias or require imputation techniques that may reduce the reliability of the analysis. The absence of missing values allows for direct comparisons without more adjustments.

Consistency was addressed through minor preprocessing steps, including the standardization of text fields such as county and agency names. This step ensured that records aligned correctly during dataset integration, preventing mismatches that could compromise the analysis.

Relevance was proven by selecting variables supporting the project's aims. Ridership measures demand, operating expenses reflect financial investment, and population provides context for evaluating use. Together, these variables form a cohesive dataset that aligns closely with the problem being analyzed.

Data Preparation and Preprocessing

Data preprocessing played a critical role in transforming raw data into a structured format suitable for analysis. Several key steps were implemented to ensure the data was clean, consistent, and ready for integration and visualization.

The first step involved cleaning and standardizing text fields. Inconsistent naming conventions, such as variations in county or agency names, can lead to errors during data merging. By standardizing these fields, the integration process was made more reliable and correct.

Next, data type conversion was performed to ensure that key variables such as ridership and operating expenses were represented as numeric values. This step was essential for enabling mathematical operations, such as calculating derived metrics and generating visualizations.

A thorough inspection of the datasets confirmed that there were no missing values. This cuts the need for imputation and ensures that all calculations were based on complete and right data.

The datasets were then merged using a common geographic identifier, creating a unified dataset that combines operational metrics with population data. This integration enabled cross-variable analysis and provided the foundation for deeper insights.

To enhance the analytical depth of the project, several derived metrics were created. Expense per trip was calculated to evaluate cost efficiency, while trips per resident measured service use compared to the population. Also, ridership and operating expenses were scaled into millions to improve readability and comparability in visualizations.

These preprocessing steps were essential for ensuring that the data was both exact and meaningful, allowing for a more robust and insightful analysis.

Exploratory Data Analysis

Exploratory data analysis revealed significant variation in scale, efficiency, and use across the three transit agencies. King County Metro works at a substantially larger scale than the other agencies, with approximately 78.9 million annual trips and over \$928 million in operating expenses. Spokane Transit Authority is at a mid-sized level, with approximately 9.4 million trips and \$108 million in expenses. In contrast, the City of Longview runs on a much smaller scale, with only around 274,000 trips and \$5.9 million in expenses.

When these values are examined compared to population, more insights appear. King County Metro achieves approximately 33.7 trips per resident, showing an elevated level of transit use. Spokane Transit Authority records approximately 16.9 trips per resident, reflecting moderate usage. Longview, however, shows only about 2.4 trips per resident, suggesting that transit plays a much smaller role in daily mobility for its population.

Efficiency also varies significantly across agencies. Both King County Metro and Spokane Transit Authority support similar expense-per-trip values, approximately 11.5 to 11.8 dollars per trip. This shows relatively strong cost efficiency at both large and mid-sized scales. In contrast, Longview's expense per trip is approximately 21.4 dollars, nearly double that of the other agencies. This suggests that smaller systems may face higher per-unit costs due to fixed operational expenses being distributed across fewer riders.

These findings highlight the importance of scale in transit operations. Larger systems receive help from economies of scale, allowing them to serve more riders at a lower cost per trip.

However, population alone does not fully explain differences in performance. The variation in trips per resident shows that factors such as service availability, infrastructure, and commuter behavior also play a significant role in finding ridership levels.

High-Level Visualization: Executive Perspective

The high-level visualization uses a bar chart to compare annual ridership across the three agencies. This visualization is designed to provide a clear and immediate overview of system performance, making it well-suited for executive stakeholders who require concise and actionable insights.

The bar chart highlights the significant disparity in ridership between the agencies. King County Metro's ridership is substantially higher than that of Spokane Transit Authority and the City of Longview, reflecting its larger population and more extensive transit network. This difference is not incremental but exponential, emphasizing the importance of scale in transit operations.

For executive decision-makers, this visualization provides a solid foundation for strategic planning. It supports decisions related to budget allocation, infrastructure investment, and service expansion by clearly finding where demand is highest. By focusing on a single key metric, the chart minimizes complexity and allows for rapid interpretation, which is essential in high-level decision-making contexts.

Detailed Visualization: Analyst Perspective

The detailed visualization uses a scatter plot with bubble sizes to integrate multiple variables into a single, comprehensive view. Population is represented on the x-axis; ridership on the y-axis, and operating expenses are reflected through bubble size.

This visualization reveals a clear positive relationship between population and ridership, showing that larger populations generally support higher levels of transit usage. However, it also highlights significant differences in efficiency and use across agencies.

The size of the bubbles illustrates how operating expenses scale with system size. Larger systems require greater financial investment, but they also serve more riders, which can improve cost efficiency. Smaller systems, while less expensive overall, may experience higher costs per trip due to lower ridership levels.

This visualization is particularly valuable for transit planners and analysts because it allows them to examine multiple variables simultaneously. It supports more detailed analysis of efficiency, cost management, and service optimization, enabling stakeholders to identify areas for improvement and make more informed operational decisions.

Stakeholder-Focused Insights

Executive Leadership

Executive stakeholders need high-level insights that support strategic decision-making. The analysis provides several key takeaways:

- Larger populations generate higher transit demand, requiring greater investment in infrastructure and services
- High ridership systems receive help from economies of scale, resulting in improved efficiency
- Resource allocation should be aligned with demand patterns to maximize impact and avoid inefficiencies

These insights support decisions related to long-term planning, budget allocation, and system expansion.

Transit Planners and Analysts

Operational stakeholders require more detailed insights to improve system performance. The analysis highlights several important considerations:

- Efficiency varies significantly across agencies, particularly in terms of cost per trip
- Smaller systems may face challenges related to fixed costs and lower use
- Population alone does not find ridership; more factors such as service design and infrastructure must be considered

These findings support decisions related to route optimization, scheduling, and cost management, enabling planners to improve both efficiency and service quality.

Effectiveness of Visualizations

Effective data visualization requires aligning design with audience needs and reducing cognitive load (Knafllic, 2015). The bar chart provides a clear and concise overview, making it ideal for executive stakeholders. The scatter plot offers a more detailed and multi-dimensional perspective, supporting deeper analysis for technical stakeholders.

Customization techniques, including clear labeling, proper scaling, and thoughtful layout design, enhance readability and ensure that the visualizations accurately stand for the data. By tailoring the complexity of each visualization to its intended audience, the analysis improves both accessibility and interpretability.

The inclusion of a summary table further strengthens the analysis by providing exact values that complement the visualizations. This allows stakeholders to confirm insights and explore the data in greater detail.

Conclusion

This project proves the power of data analysis and visualization in transforming raw data into actionable insights. By integrating transit performance data with population data, the analysis reveals meaningful patterns in ridership, efficiency, and resource allocation.

The findings show that while population size is a major driver of transit demand, it does not fully explain differences in system performance. Larger systems receive help from economies of scale, while smaller systems face challenges related to cost efficiency and use.

The use of multi-level visualizations ensures that insights are accessible to both strategic and operational stakeholders, highlighting the importance of tailoring communication to different audiences.

Ultimately, this analysis reinforces the value of data-driven decision-making in improving public transportation systems and supporting more efficient and effective resource allocation. These findings highlight the importance of using data visualization and storytelling techniques to support decision-making (Knafllic, 2015).

References

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